
An Information-Theoretic Evaluation Framework for Benchmark and Model Diagnosis in Knowledge Tracing

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Abstract

1 Knowledge tracing (KT) models are predominantly evaluated using aggregate
2 metrics such as area under the curve (AUC) and accuracy. However, these global
3 scores obscure where the remaining errors originate and fail to indicate whether
4 a benchmark is approaching saturation. While estimating a global theoretical
5 performance limit is challenging in realistic KT settings, it is possible to quantify
6 local predictability. To address this, we propose an information-theoretic evaluation
7 framework for KT benchmark diagnosis. We utilize Context Tree Weighting (CTW)
8 as an operational causal anchor to estimate the Local Irreducible Uncertainty (LIU)
9 of each student interaction. By projecting predictions onto this shared uncertainty
10 coordinate, we evaluate model performance gains across distinct entropy bands
11 rather than only at the global level. Comprehensive evaluations on NIPS Task 3/4
12 and Algebra 2005 reveal that model improvements are highly non-uniform. While
13 the largest gains achieved by modern KT models consistently occur in high-entropy
14 regions, indicating substantial remaining benchmark headroom, our framework also
15 identifies instances where models capture inherently random noise. By surfacing
16 these local modeling failures alongside genuine gains, this approach provides a
17 rigorous diagnostic tool to pinpoint both the potential and the limitations of current
18 KT benchmark and model diagnosis.

19 1 Introduction

20 Knowledge tracing (KT) predicts a student’s future responses from their past interaction history,
21 serving as a foundational task in artificial intelligence (AI) for education. As modern KT architectures
22 grow increasingly sophisticated, performance on standard benchmarks continues to climb. However,
23 the evaluation paradigm remains dominated by aggregate metrics such as area under the curve
24 (AUC) and accuracy. These global scores are useful for leaderboard rankings, but they fail to answer
25 two diagnostic questions that matter for progress: *where does meaningful benchmark headroom*
26 *remain?* and *what kind of errors is a model reducing?* A marginal +0.01 boost in global AUC
27 cannot distinguish between a model that refines already-easy cases, one that extracts genuine structure
28 from difficult interactions, and one whose apparent gain is concentrated in unstable regions where
29 outcomes are close to random under the observable protocol.

30 To address this gap, we propose an *information-theoretic* evaluation framework for KT benchmark
31 and model diagnosis. The framework separates two roles that aggregate evaluation conflates. First,
32 **uncertainty characterization** establishes a shared local coordinate for prediction difficulty. Because
33 the true Local Irreducible Uncertainty (LIU) is not directly observable in finite KT data, we use
34 Context Tree Weighting (CTW) as an operational causal anchor: it assigns each prediction point a
35 fold-consistent entropy estimate using only past response history. Second, **predictive realization**

measures how much task-facing KT models and anchor-side neural probes improve on that coordinate. This decoupling lets us ask not only whether a benchmark is moving on average, but also whether gains occur in low-uncertainty regions, high-uncertainty regions with exploitable structure, or high-uncertainty regions that call for item or data inspection.

Our contributions are as follows: (i) We introduce an information-theoretic evaluation paradigm that shifts KT assessment from global leaderboard scores to localized benchmark and model diagnosis; (ii) We formalize the separation between local irreducible uncertainty and realized predictive performance, with CTW serving as an operational causal anchor rather than as the ground-truth LIU; (iii) We provide a fold-consistent empirical protocol based on CTW-anchored entropy bands, gain-over-anchor curves, and item-level remaining-uncertainty/information-gain diagnostics; and (iv) Through broad aligned experiments on NIPS Task 3/4 and Algebra 2005, we show that the largest reliable gains occur in high-entropy regions while also identifying high-uncertainty, low-information-gain cases where apparent model improvements should be treated as noise-sensitive behavior rather than unambiguous progress.

2 Related Work

KT has progressed from Bayesian KT [Corbett and Anderson, 1994] to neural sequence models such as DKT [Piech et al., 2015], memory-based models such as DKVMN [Zhang et al., 2017], attention-based variants such as SAKT and AKT [Pandey and Karypis, 2019, Ghosh et al., 2020], and strong modern baselines such as Deep-IRT and simpleKT [Yeung, 2019, Liu et al., 2023b]. Toolkits and benchmarks including ASSISTments, EdNet, the NeurIPS 2020 Education Challenge, and pyKT have improved reproducibility [Heffernan and Heffernan, 2014, Choi et al., 2020b, Wang et al., 2021b, Liu et al., 2022]. However, these comparisons still primarily report aggregate metrics such as AUC and accuracy.

Our work is closer to probabilistic forecast evaluation. Information-theoretic uncertainty [Cover and Thomas, 2005], universal coding tools such as CTW [Willems et al., 1995], and proper scoring rules such as Brier score and log-loss [Brier, 1950, Gneiting and Raftery, 2007] provide tools for separating uncertainty from realized predictive quality. We use CTW as a common causal anchor and evaluate model gains along entropy bands rather than only at the benchmark level. This makes the contribution diagnostic rather than architectural: the objective is to determine where benchmark progress is reliable, where model capacity changes predictions, and where apparent gains are more consistent with unstable or noise-like item behavior.

3 Framework

3.1 Problem setup and local uncertainty

In standard KT, we observe a student’s interaction history up to time t , denoted as $X_t = \{(q_1, r_1), \dots, (q_t, r_t)\}$, where q_k represents the practiced item and $r_k \in \{0, 1\}$ represents the correctness of the response. Given the next item q_{t+1} , the task is to predict the probability of a correct response:

$$\hat{p}_{t+1} = \hat{P}(R_{t+1} = 1 \mid X_t, q_{t+1}). \quad (1)$$

To formalize the information available at this prediction step, we define the observable filtration as $\mathcal{F}_t = \sigma(X_t, q_{t+1})$. Based on this, we define the *local irreducible uncertainty* (LIU) of the prediction point as the conditional entropy of the true response R_{t+1} :

$$\text{LIU}(X_t, q_{t+1}) = H(R_{t+1} \mid \mathcal{F}_t). \quad (2)$$

This term represents the theoretical entropy remaining under the admissible KT information boundary, i.e., the inherent unpredictability even if we had a perfect model for the student’s historical data. In finite benchmark data, LIU is not directly observed; it must be studied through operational causal estimates that respect the information available at prediction time.

Rather than aggregating all predictions into a single global metric, our framework evaluates models locally. For each prediction instance i across the dataset, we compute an operational CTW uncertainty coordinate $\hat{u}_{\text{ctw},i}$ as a causal proxy for LIU_i , partition the samples into distinct uncertainty buckets $\mathcal{B}_k$, and evaluate the model’s performance $\text{Perf}_m(k)$ within each specific bucket.

Uncertainty-Conditioned Evaluation Framework for KT Diagnosis

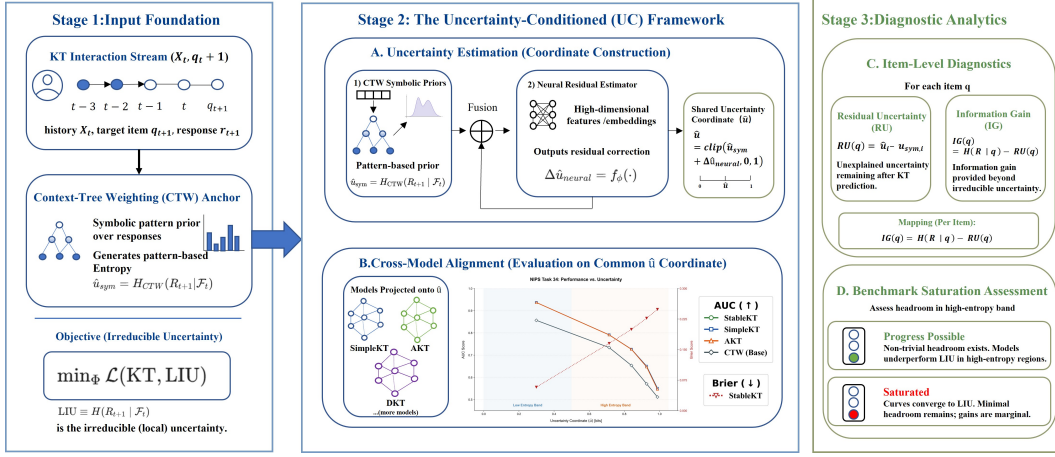


Figure 1: Overview of the proposed information-theoretic evaluation framework. A causal anchor assigns each prediction point to a shared uncertainty coordinate; model performance is then read along this coordinate to reveal where benchmark headroom remains and where model behavior warrants further diagnosis.

3.2 Entropy-risk decomposition and causal anchor

Let $P_t = P(R_{t+1} | \mathcal{F}_t)$ be the true conditional response distribution. For any $\mathcal{F}_t$ -measurable probabilistic predictor $\hat{P}_\theta$, the expected log-loss (cross-entropy) naturally decomposes into two components:

Fact 1 (Local entropy-risk decomposition).

$$\mathbb{E}[-\log \hat{P}_\theta(R_{t+1} | \mathcal{F}_t)] = \underbrace{H(R_{t+1} | \mathcal{F}_t)}_{\text{LIU}} + \underbrace{\mathbb{E}[\text{KL}(P_t \| \hat{P}_\theta)]}_{\text{epistemic gap}}. \quad (3)$$

This decomposition illustrates that standard benchmark-level error inextricably mixes LIU with model mismatch (the epistemic gap). A global score can decrease either because a model reduces epistemic error in a meaningful region or because gains and failures cancel across regions with very different local uncertainty.

To separate these components operationally, we use Context Tree Weighting (CTW) as an autoregressive causal anchor. For each prediction point, CTW maps the historical binary response prefix to a next-response probability p_{ctw} using strictly past information. The corresponding uncertainty coordinate is its entropy: $\hat{u}_{\text{ctw},t} = H(p_{\text{ctw}})$. We use CTW as a reproducible information-theoretic coordinate. Its entropy, $H(p_{\text{ctw}})$, acts as an empirical baseline to expose exploitable data structures wherever stronger predictors outperform it in high-entropy regions.

Furthermore, we introduce a fused CTW+NN predictor. This model augments the pure CTW anchor by applying a neural refinement of the form $p_{\text{ctw}} \mapsto \sigma(\text{logit}(p_{\text{ctw}}) + f_\phi(X_t, q_{t+1}))$. In our evaluation protocol, the pure CTW model defines the shared uncertainty axis (i.e., the buckets), while the CTW+NN model acts as an anchor-side neural probe. It shows whether exploitable structural signals still exist beyond the capabilities of the symbolic source model.

3.3 Operational bands, remaining uncertainty, and information gain

Given an aligned evaluation set, our empirical protocol proceeds as follows: First, we compute a fold-consistent CTW prediction $p_{\text{ctw},i}$ for each evaluation point i . Second, we convert this probability into an estimated entropy score $\hat{u}_{\text{ctw},i} = H(p_{\text{ctw},i})$. Third, we partition the samples into distinct uncertainty buckets (using either equal-width or equal-frequency binning). Finally, we report both the absolute performance and the performance gain over the CTW anchor within each bucket. This

109 resulting band view is deliberately model-agnostic: any KT baseline can be evaluated on this exact
110 same CTW-defined coordinate.

111 The same autoregressive anchor also enables fine-grained, item-level diagnostics. For any given item
112 q , let H_q^{marg} denote the entropy of its marginal empirical correctness, and let H_q^{ctw} denote its average
113 CTW conditional entropy across all predictions for that item. We define the *remaining uncertainty*
114 as $\text{RU}(q) = H_q^{\text{ctw}}$ and the *information gain* as $\text{IG}(q) = H_q^{\text{marg}} - H_q^{\text{ctw}}$. A high RU identifies
115 items whose outcomes remain unstable under the current observable protocol (e.g., potentially due to
116 guessing, ambiguous wording, missing observables, or overly coarse step granularity); conversely,
117 a high IG highlights items where knowing a student’s historical context substantially reduces the
118 prediction uncertainty.

119 3.4 Benchmark and model diagnosis criteria

120 The framework answers three diagnostic questions. First, **benchmark diagnosis** asks whether
121 high-uncertainty regions with substantial support still admit reliable gains over the causal anchor; if
122 strong models remain separated there, aggregate saturation claims are premature. Second, **model**
123 **diagnosis** asks where a model improves and whether those improvements are supported by proper
124 scoring behavior, not only by threshold accuracy. Third, **noise-sensitive behavior** asks whether
125 apparent improvements occur in high-RU/low-IG regions where observable history provides little
126 information. Such movements should not be interpreted as progress unless they are stable across
127 folds and accompanied by proper-score improvements.

128 4 Experimental Protocol

129 We evaluate five-fold aligned exports on NIPS Task 3/4 and Algebra 2005. ASSIST2015 is reported
130 as a global auxiliary check, while the local diagnostic figures focus on NIPS Task 3/4 and Algebra
131 2005 because these datasets support the band-level and item-level diagnostics used in the main
132 figures. Each prediction point is assigned a CTW entropy value and placed into one of five entropy
133 bands $[0.0, 0.2)$, $[0.2, 0.4)$, $[0.4, 0.6)$, $[0.6, 0.8)$, $[0.8, 1.0]$. We report AUC, accuracy, and Brier score
134 globally and within bands. Accuracy gives the decision-level view, while Brier score is used as a
135 proper-score check on whether probability quality improves in the same region.

136 The model pool includes the CTW anchor, the CTW+NN neural probe, and a broad KT sweep
137 spanning DKT, DKT+, DKT-forget, KQN, SAKT, SAINT, AKT, simpleKT, sparseKT, FoLiBiKT,
138 stableKT, extraKT, Deep-IRT, ATKKT, ATKTFix, DIMKT, ATDKT, and HawkesKT [Yeung and
139 Yeung, 2018, Nagatani et al., 2019, Lee and Yeung, 2019, Choi et al., 2020a, Guo et al., 2021, Huang
140 et al., 2023, Im et al., 2023, Li et al., 2024a,b, Shen et al., 2022, Liu et al., 2023a, Wang et al., 2021a].
141 The main bucket coordinate is always computed from CTW alone, so task-facing KT baselines do not
142 define the axis on which they are judged. Model diagnosis is then based on the alignment between
143 gain-over-anchor curves, proper-score changes, fold consistency, and item-level RU/IG signals.

144 5 Results

145 5.1 Sanity check: same marginal correctness, different causal uncertainty

146 Before real benchmarks, we verify the property needed from the anchor. We construct two binary
147 response regimes with nearly identical marginal correctness rates. The first is an i.i.d. Bernoulli(0.5)
148 process. The second is a symmetric persistent two-state Markov process: the initial state is uniform
149 over $\{0, 1\}$, and each subsequent response copies the previous response with probability 0.95 and
150 flips with probability 0.05. This process has stationary marginal correctness 0.5 but forms long
151 correct/incorrect runs. Across 64 sequences of length 400, the observed correctness rates are 0.492
152 and 0.495, yet average CTW entropy is 0.986 bits for the i.i.d. regime and 0.318 bits for the persistent
153 regime. CTW predictive accuracy is 0.501 versus 0.945.

154 5.2 Global aligned performance is insufficient for diagnosis

155 Global aligned performance is used as context rather than as the main diagnostic target. Across the
156 aligned folds, CTW+NN improves the CTW anchor on all three datasets while remaining an anchor-

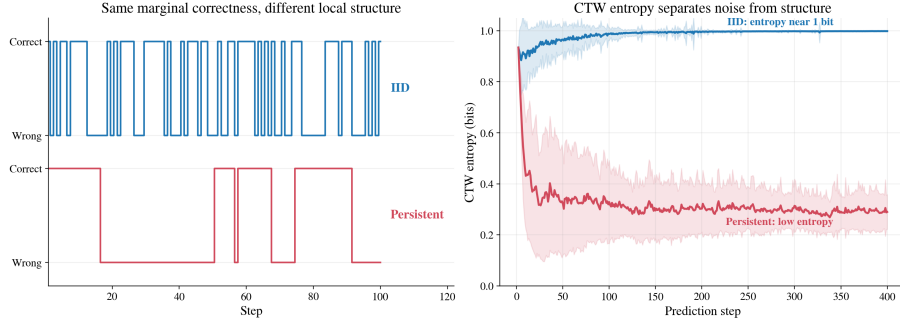


Figure 2: Synthetic sanity check. Two regimes have nearly identical marginal correctness near 0.5, but CTW separates noise-like responses from run-structured responses with strong causal predictability.

Table 1: Task-facing KT baselines across the three benchmark datasets. Each metric cell is formatted as a two-line entry with the mean emphasized and the standard deviation reduced for readability. Entries are mean $\pm$ standard deviation over five folds; lower Brier is better. Best entries are bolded and second-best entries are underlined within each dataset/metric block. The unified table keeps only the KT models available on all three datasets. All NIPS Task 3/4 rows use the shared aligned subset of 1,354,735 points. For Algebra 2005, all rows use 464,725 points except ATKTFix (271,122) and ATKTFix (278,835). For ASSIST2015 under the convention *concept* = *question*, all rows use 672,960 points.

Model	NIPS Task 3/4			Algebra 2005			ASSIST2015		
	AUC	ACC	Brier	AUC	ACC	Brier	AUC	ACC	Brier
simpleKT	0.8014 ± 0.0003	<u>0.7309</u> ± 0.0004	<u>0.1798</u> ± 0.0005	0.8147 ± 0.0225	0.8065 ± 0.0072	0.1389 ± 0.0060	0.7231 ± 0.0012	0.7502 ± 0.0006	0.1713 ± 0.0003
stableKT	<u>0.8011</u> ± 0.0005	0.7310 ± 0.0003	0.1796 ± 0.0003	0.8200 ± 0.0181	0.8070 ± 0.0063	0.1365 ± 0.0052	<u>0.7256</u> ± 0.0010	<u>0.7506</u> ± 0.0002	<u>0.1707</u> ± 0.0002
AKT	0.7999 ± 0.0005	0.7292 ± 0.0008	0.1805 ± 0.0004	<u>0.8277</u> ± 0.0015	<u>0.8110</u> ± 0.0015	<u>0.1334</u> ± 0.0007	0.7262 ± 0.0008	0.7519 ± 0.0008	0.1708 ± 0.0002
FoLiBiKT	0.7993 ± 0.0008	0.7291 ± 0.0010	0.1809 ± 0.0002	0.8026 ± 0.0141	0.8017 ± 0.0049	0.1413 ± 0.0043	0.7261 ± 0.0013	<u>0.7520</u> ± 0.0004	0.1707 ± 0.0004
extraKT	0.7982 ± 0.0020	0.7285 ± 0.0016	0.1810 ± 0.0011	0.8215 ± 0.0029	0.8064 ± 0.0018	0.1365 ± 0.0011	0.7257 ± 0.0004	0.7506 ± 0.0002	0.1712 ± 0.0001
sparseKT-soft	0.7979 ± 0.0002	0.7276 ± 0.0003	0.1813 ± 0.0002	0.8155 ± 0.0009	0.8031 ± 0.0011	0.1395 ± 0.0006	0.7206 ± 0.0013	0.7494 ± 0.0004	0.1718 ± 0.0003
sparseKT-topk	0.7974 ± 0.0010	0.7271 ± 0.0008	0.1814 ± 0.0004	0.8137 ± 0.0012	0.8020 ± 0.0009	0.1407 ± 0.0002	0.7161 ± 0.0066	0.7490 ± 0.0024	0.1728 ± 0.0016
ATKTFix	0.7776 ± 0.0062	0.7086 ± 0.0045	0.1904 ± 0.0026	0.8373 ± 0.0229	0.8179 ± 0.0189	0.1280 ± 0.0128	0.8069 ± 0.0100	0.7787 ± 0.0062	0.1494 ± 0.0035
DKT+	0.7660 ± 0.0006	0.7010 ± 0.0007	0.1947 ± 0.0003	0.8098 ± 0.0016	0.8078 ± 0.0006	0.1372 ± 0.0005	<u>0.7283</u> ± 0.0007	0.7510 ± 0.0004	<u>0.1702</u> ± 0.0002
DKT	0.7659 ± 0.0002	0.7005 ± 0.0004	0.1948 ± 0.0001	0.8091 ± 0.0007	0.8072 ± 0.0005	0.1374 ± 0.0004	0.7267 ± 0.0005	0.7506 ± 0.0006	0.1706 ± 0.0001
KQN	0.7626 ± 0.0011	0.6982 ± 0.0006	0.1961 ± 0.0005	0.7883 ± 0.0062	0.7985 ± 0.0020	0.1437 ± 0.0017	0.7250 ± 0.0007	0.7500 ± 0.0003	0.1709 ± 0.0001
SAINT	0.7621 ± 0.0303	0.6964 ± 0.0258	0.1975 ± 0.0105	0.7660 ± 0.0133	0.7780 ± 0.0031	0.1607 ± 0.0039	0.7007 ± 0.0020	0.7441 ± 0.0015	0.1765 ± 0.0009
Deep-IRT	0.7599 ± 0.0029	0.6951 ± 0.0025	0.1971 ± 0.0013	0.7949 ± 0.0020	0.8006 ± 0.0007	0.1421 ± 0.0005	0.7185 ± 0.0003	0.7501 ± 0.0004	0.1719 ± 0.0001
ATKTFix	0.7599 ± 0.0032	0.6956 ± 0.0024	0.1973 ± 0.0012	0.7841 ± 0.0051	0.7941 ± 0.0035	0.1453 ± 0.0017	0.7213 ± 0.0035	0.7485 ± 0.0013	0.1719 ± 0.0008
SAKT	0.7481 ± 0.0004	0.6844 ± 0.0010	0.2031 ± 0.0011	0.7867 ± 0.0037	0.7964 ± 0.0016	0.1477 ± 0.0022	0.7112 ± 0.0004	0.7472 ± 0.0001	0.1756 ± 0.0007

side neural probe rather than the definition of the uncertainty axis: +0.0823 AUC and +0.0690 accuracy on NIPS Task 3/4, +0.0835 AUC and +0.0160 accuracy on Algebra 2005, and +0.0455 AUC and +0.0142 accuracy on ASSIST2015. The task-facing KT sweep is reported in Table 1. These numbers show that stronger predictors improve on average, but they do not reveal whether the improvement comes from already predictable regions, genuinely difficult but structured regions, or unstable regions where apparent gains require caution. The central claim is local: global scores provide a sanity check, while the entropy-conditioned bands below test where benchmark headroom and model limitations actually lie.

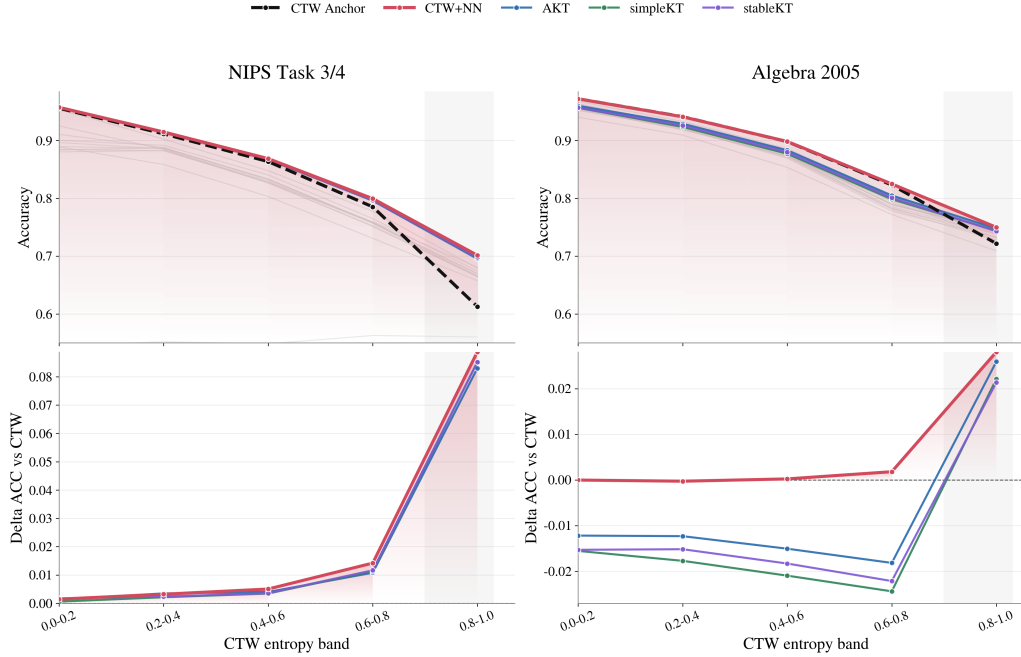


Figure 3: Entropy-conditioned accuracy bands under the CTW anchor. Top: absolute accuracy by CTW entropy band. Bottom: accuracy gain relative to CTW. Higher CTW-entropy bands have lower accuracy, but the largest gains over CTW also occur in these highest-entropy regions, indicating difficult but not exhausted benchmark regions.

Table 2: CTW+NN minus CTW improvement by CTW entropy band. Positive ΔACC and negative ΔBrier indicate gains over the anchor.

Band	NIPS Task 3/4		Algebra 2005	
	ΔACC	ΔBrier	ΔACC	ΔBrier
[0.0, 0.2)	+0.0026	−0.0075	−0.0002	−0.0006
[0.2, 0.4)	+0.0037	−0.0144	+0.0002	−0.0019
[0.4, 0.6)	+0.0055	−0.0161	−0.0001	−0.0041
[0.6, 0.8)	+0.0147	−0.0229	+0.0021	−0.0087
[0.8, 1.0]	+0.0886	−0.0378	+0.0283	−0.0681

5.3 Benchmark diagnosis through entropy bands

The main empirical pattern is local rather than global. On NIPS Task 3/4, the accuracy improvement of CTW+NN over CTW grows from +0.0026 in the lowest entropy band to +0.0886 in the highest. On Algebra 2005, gains are small in lower bands but reach +0.0283 accuracy and 0.0681 Brier reduction in the highest band. Moreover, the highest band is not a tiny tail: it contains roughly 74.8% of NIPS Task 3/4 samples and 55.6% of Algebra 2005 samples in the shared aligned subsets. These high-entropy regions are therefore central to the benchmark rather than peripheral outliers.

5.4 Model diagnosis through gain profiles

The band profile should be read as a diagnostic object rather than as a replacement for global model ranking. The absolute-accuracy row in Figure 3 answers a first-order question: does the CTW coordinate actually stratify prediction difficulty? In both main datasets, the answer is yes. Accuracy falls as CTW entropy increases, and this trend is visible not only for the CTW anchor but also for the KT baselines. This matters because the coordinate is computed without access to future labels or to the task-facing KT model being evaluated. It therefore provides a common, causal axis on which different predictors can be compared.

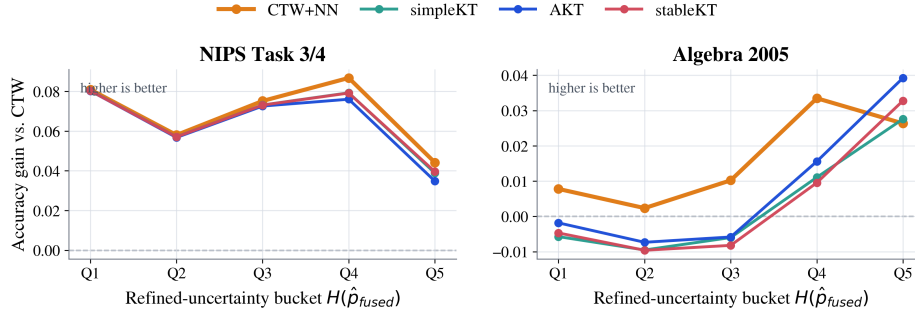


Figure 4: Sensitivity analysis using the refined uncertainty coordinate $\hat{u}_{refined} = H(\hat{p}_{fused})$. The compact view focuses on accuracy gains relative to the CTW anchor, the primary headroom signal used in the band analysis. The refined coordinate preserves a non-uniform gain profile while avoiding the visual clutter of duplicating absolute-metric curves.

180 The delta row answers a second-order question: where is there still exploitable headroom relative
 181 to the anchor? If a benchmark were nearly exhausted under the current observable protocol, then
 182 high-entropy regions should show little remaining separation among strong models. That is not the
 183 observed pattern. On NIPS Task 3/4, the CTW+NN neural probe barely moves the lowest-entropy
 184 band but gains almost nine accuracy points in the highest band. The main KT baselines follow the
 185 same qualitative profile. This shows that the difficult region is not merely random noise: it is difficult
 186 for the symbolic anchor, but still contains structure that stronger predictors can exploit.

187 Algebra 2005 shows a weaker low-band pattern but the same high-band conclusion. In the first four
 188 entropy bands, CTW+NN is close to CTW in accuracy, while several KT baselines can sit below the
 189 anchor depending on coverage and model family. In the highest band, however, CTW+NN separates
 190 clearly, and the Brier reduction is largest. This is exactly the regime where a local diagnostic is
 191 more informative than a global score: aggregate performance averages together easy regions that are
 192 already largely solved and high-uncertainty regions where model choice still matters.

193 This interpretation also clarifies why CTW+NN is included. It is not used to define the benchmark
 194 axis, and it is not presented as a new KT leaderboard method. Its role is to act as an anchor-side neural
 195 probe: if a simple neural refinement of the causal anchor improves most in high-entropy regions, then
 196 those regions should not be dismissed as irreducible noise. They are hard, but not exhausted. By
 197 contrast, flat delta curves in high-entropy bands across strong models would indicate a more saturated
 198 benchmark under the same observable protocol.

199 The same profile is also a model-diagnosis tool, but the roles must remain separate: CTW+NN is
 200 an anchor-side probe for remaining signal, not a task-facing KT baseline. For task-facing models, a
 201 gain is most credible when threshold accuracy and Brier score move in the same direction within
 202 the same region. Isolated accuracy changes in high-entropy regions should be inspected rather
 203 than immediately counted as progress, because high entropy can arise from both exploitable hidden
 204 structure and effectively random variation under the observed features. This is why the framework
 205 pairs band-level gains with item-level RU/IG diagnostics: the band profile identifies where a model
 206 changes behavior, while RU/IG asks whether that region contains contextual signal or unstable item
 207 behavior.

208 5.5 Sensitivity and noise-sensitive item diagnostics

209 Two additional checks support the same interpretation. First, repartitioning samples by a refined
 210 uncertainty coordinate $\hat{u}_{refined} = H(\hat{p}_{fused})$ preserves the non-uniform headroom profile, while
 211 smoothing the NIPS gain curve because the fused predictor already absorbs part of the structure
 212 left unresolved by CTW. On Algebra 2005, higher refined-uncertainty buckets still exhibit larger
 213 accuracy gains over the CTW anchor; on NIPS Task 3/4, the peak moves from the last bucket to the
 214 upper-middle region, consistent with the refined coordinate already encoding part of the additional
 215 signal.

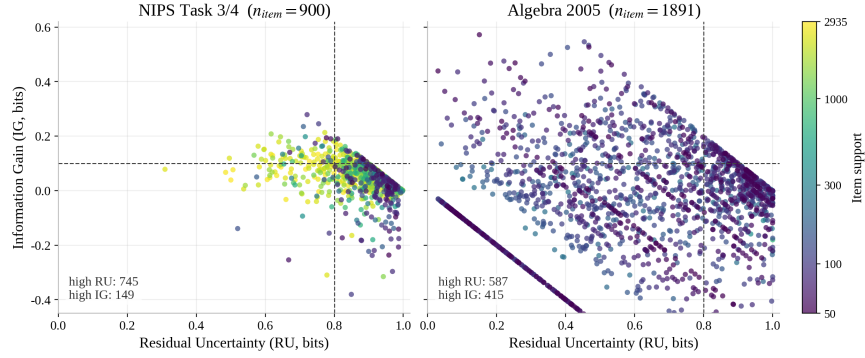


Figure 5: Item-level diagnostic map under the CTW anchor. RU captures remaining item uncertainty and IG captures contextual signal beyond item marginal correctness. High-RU/low-IG regions are the main warning cases for noise-sensitive model behavior, while high-IG regions indicate contextual structure that models can exploit.

Second, item-level RU and IG separate item regions whose outcomes remain unstable from regions where history carries strong predictive signal. This is the main mechanism by which the framework surfaces noise-sensitive model behavior. High-RU/low-IG items point to ambiguity, guessing, missing observables, or problematic granularity; if a model’s apparent improvement is concentrated in such items without proper-score or fold-stability support, the improvement is more consistent with capturing effectively random variation than with learning reliable structure. High-IG items, in contrast, point to regions where better temporal or prerequisite modeling should pay off. Figure 5 shows this support-filtered diagnostic map.

The RU/IG view makes the distinction concrete. High-RU/low-IG items have near-chance empirical correctness and CTW probabilities near 0.5, so apparent model gains on them require proper-score and fold-stability checks before being treated as progress. Figure 6 compares all task-facing baselines with a direct two-score diagnostic: the x -coordinate asks whether accuracy improves over CTW, the y -coordinate asks whether the proper score improves over CTW, and the green quadrant marks gains supported by both criteria. This makes the role of the added model-diagnosis paragraph explicit: it converts the RU/IG map from an item-level warning signal into a model-level reliability check. The strongest NIPS models cluster in the green quadrant for high-IG regions, whereas Algebra and high-RU/low-IG panels contain more leftward or low- y points. Thus uncertainty bands identify *where* headroom remains, while RU/IG distinguishes exploitable structure from unstable item behavior.

6 Discussion and Limitations

The results support information-theoretic reporting as a benchmark- and model-diagnosis tool. A benchmark close to saturation should show tight clustering of strong models especially in high-uncertainty regions. We do not observe that pattern for NIPS Task 3/4 or Algebra 2005: the hardest CTW-defined regions still contain the largest gains. This suggests that aggregate KT scores hide substantial local structure and that these benchmarks remain far from exhausted under the current observable protocol.

CTW is used here as a causal anchor and common analysis coordinate, not as a complete model of student learning. This localizes where headroom remains while avoiding the confound that would arise if every model defined its own uncertainty axis. The refined-coordinate and Brier checks reduce the risk that the conclusion is an artifact of CTW-only bucketization or threshold accuracy. At the same time, the framework avoids treating every high-entropy gain as progress: high-RU/low-IG items expose regions where model changes should be interpreted as noise-sensitive unless supported by proper-score improvement and fold stability.

The analysis has three scope boundaries. First, alternative uncertainty coordinates may reveal complementary structure. Second, some raw KT exports have dataset-specific coverage differences, so the main claims emphasize shared aligned subsets. Third, Figure 6 reports model diagnosis at

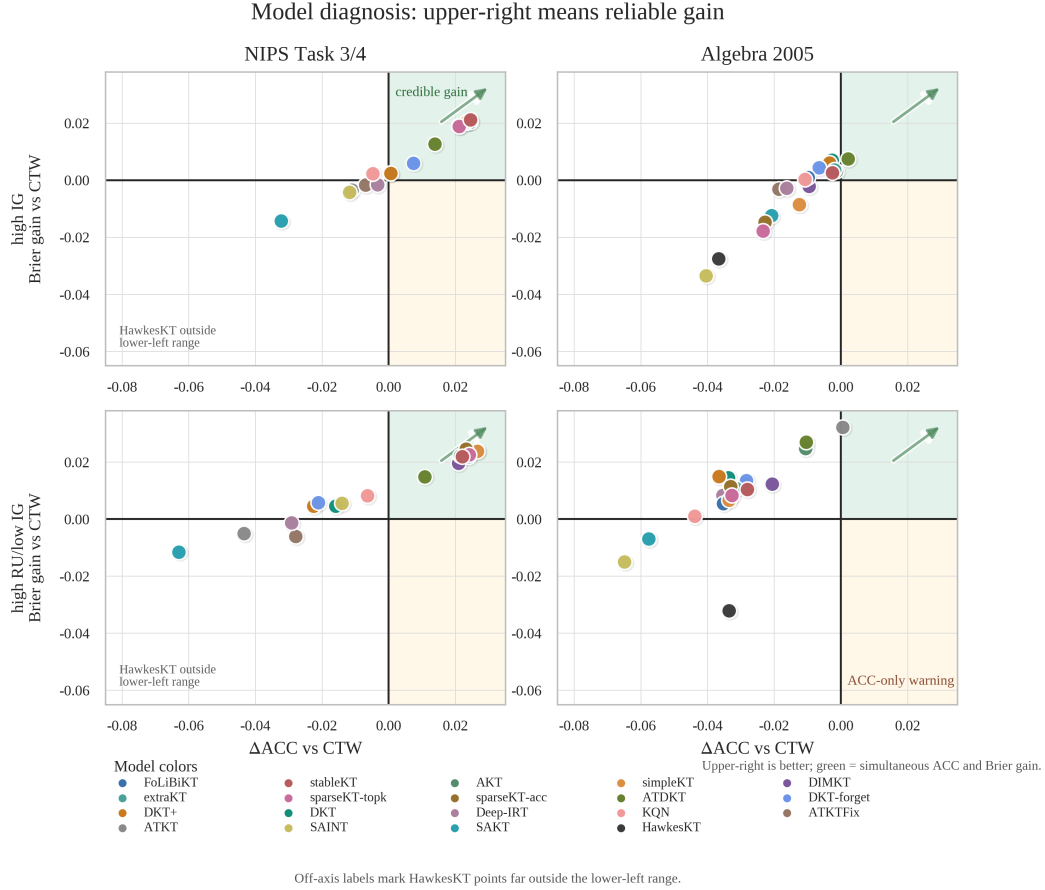


Figure 6: Model diagnosis scatter plot for task-facing KT baselines. Columns separate datasets and rows separate RU/IG regions. Each point is one KT model, with color identifying the model. The horizontal axis reports ΔACC versus CTW and the vertical axis reports Brier gain ($-\Delta\text{Brier}$) versus CTW, so farther to the upper-right is better. Points in the green upper-right quadrant improve both threshold accuracy and probability quality; points moving right but not up indicate accuracy-only gains that require caution.

the RU/IG-region level, while the paper reports item-level RU/IG summaries for NIPS Task 3/4 and Algebra 2005.

7 Conclusion

We introduced an information-theoretic evaluation framework for KT benchmark and model diagnosis. By separating uncertainty characterization from predictive realization and using CTW entropy as a shared causal coordinate, the framework shows where model progress occurs. Across broad aligned sweeps, the largest gains concentrate in high-uncertainty regions, demonstrating that these benchmarks contain substantial exploitable structure beyond what aggregate scores reveal.

The practical message is that KT benchmarks should not be judged only by aggregate score movement. A small global improvement can be meaningful if it occurs in a difficult, high-uncertainty region with substantial support and proper-score improvement; conversely, gains concentrated in high-RU/low-IG regions may reflect noise-sensitive behavior. A minimal adoption path is straightforward: report global metrics, add entropy-conditioned gain curves on a fixed causal coordinate, and inspect RU/IG regions when the curve shows remaining headroom.

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385 A Appendix A: Extended Theoretical Framework and Proofs

386 A.1 Binary entropy properties and the LIU proxy

387 For a binary correctness probability $p \in [0, 1]$, the entropy in bits is

$$H_2(p) = -p \log_2 p - (1 - p) \log_2 (1 - p). \quad (\text{A.1})$$

388 Its first two derivatives on $(0, 1)$ are

$$\frac{dH_2}{dp}(p) = \log_2 \frac{1-p}{p}, \quad \frac{d^2H_2}{dp^2}(p) = -\frac{1}{(\ln 2)p(1-p)} < 0. \quad (\text{A.2})$$

389 Equation (A.2) shows three properties that matter for the appendix. First, $H_2(p)$ is symmetric around
390 $p = 0.5$. Second, it is strictly increasing on $[0, 0.5]$ and strictly decreasing on $[0.5, 1]$. Third, it
391 is strictly concave, so it penalizes overconfident extreme probabilities more heavily than a linear
392 uncertainty score. As a result, $H_2(p)$ is small when the next response is causally predictable and
393 maximal at one bit when the best available causal probability is near chance.

394 **Fact A.1** (Local entropy-risk decomposition). *For any $\mathcal{F}_t$ -measurable probabilistic predictor $\hat{P}_\theta$,*

$$\mathbb{E}[-\log \hat{P}_\theta(R_{t+1} | \mathcal{F}_t)] = H(R_{t+1} | \mathcal{F}_t) + \mathbb{E}[\text{KL}(P_t \| \hat{P}_\theta)]. \quad (\text{A.3})$$

395 *Proof.* Let $P_t(r) = P(R_{t+1} = r | \mathcal{F}_t)$ for $r \in \{0, 1\}$. Then

$$\mathbb{E}[-\log \hat{P}_\theta(R_{t+1} | \mathcal{F}_t)] = \mathbb{E}\left[\sum_{r \in \{0,1\}} P_t(r) (-\log \hat{P}_\theta(r | \mathcal{F}_t))\right]. \quad (\text{A.4})$$

396 Adding and subtracting $\log P_t(r)$ inside the sum yields

$$\mathbb{E}\left[\sum_r P_t(r) (-\log P_t(r))\right] + \mathbb{E}\left[\sum_r P_t(r) \log \frac{P_t(r)}{\hat{P}_\theta(r | \mathcal{F}_t)}\right]. \quad (\text{A.5})$$

397 The first term in (A.5) is $H(R_{t+1} | \mathcal{F}_t)$ and the second term is $\mathbb{E}[\text{KL}(P_t \| \hat{P}_\theta)]$, which proves (A.3).
398 $\square$

399 **Fact A.2** (Binary Bayes error lower bound). *Let $P_e^*(t)$ be the Bayes 0-1 error for predicting R_{t+1}*
400 *from $\mathcal{F}_t$. Then*

$$P_e^*(t) \geq h_2^{-1}(H(R_{t+1} | \mathcal{F}_t)), \quad (\text{A.6})$$

401 *where $h_2(p) = -p \log_2 p - (1 - p) \log_2 (1 - p)$ and h_2^{-1} denotes its inverse on $[0, 1/2]$.*

Algorithm 1 CTW-based uncertainty estimation for KT evaluation

- 1: **Input:** Binary response stream $R = (r_1, \dots, r_T)$, maximum depth D
 - 2: Initialize a depth- D context tree with KT priors at every node
 - 3: **for** $t = 0$ **to** $T - 1$ **do**
 - 4: Traverse the suffix path induced by the observed prefix $r_{1:t}$
 - 5: Compute the root mixture probability $\hat{p}_{\text{ctw},t}$ by the CTW recursion in (A.10) and the predictive ratio in (A.11)
 - 6: Convert the anchor probability into uncertainty: $\hat{u}_{\text{ctw},t} \leftarrow H_2(\hat{p}_{\text{ctw},t})$
 - 7: Observe r_{t+1} and update the node statistics on the same suffix path
 - 8: **end for**
 - 9: **Output:** Anchor probabilities $\{\hat{p}_{\text{ctw},t}\}_{t=0}^{T-1}$ and uncertainty profile $\{\hat{u}_{\text{ctw},t}\}_{t=0}^{T-1}$
-

402 *Proof.* For binary targets, the conditional form of Fano’s inequality gives

$$H(R_{t+1} \mid \mathcal{F}_t) \leq h_2(P_e^*(t)). \quad (\text{A.7})$$

403 Because $P_e^*(t) \in [0, 1/2]$ and h_2 is strictly increasing on that interval by (A.2), applying h_2^{-1} to (A.7)
 404 yields (A.6). $\square$

405 These facts explain why the appendix uses

$$\hat{u}_{\text{ctw},t} = H_2(\hat{p}_{\text{ctw},t}) \quad (\text{A.8})$$

406 as the operational LIU proxy. The quantity in (A.8) depends only on the one-step-ahead Bernoulli
 407 law implied by the causal anchor, it is invariant to threshold choices, and it is directly aligned with
 408 the cross-entropy decomposition in (A.3).

409 A.2 CTW algorithm details and cold-start behavior

410 Let $R = (r_1, \dots, r_T)$ denote the pure response sequence extracted from the original interaction
 411 sequence $X_T = \{(q_1, r_1), \dots, (q_T, r_T)\}$. The CTW anchor is defined on this binary response stream
 412 and uses only the observable response prefix. Let D be the maximum context depth and let λ denote
 413 the empty context. For each context s with $|s| \leq D$, let r^s be the subsequence of responses whose
 414 preceding suffix equals s . The Krichevsky-Trofimov block probability at node s is

$$P_e^s(r^s) = \prod_{j=1}^{|r^s|} \frac{n_{s,j-1}^{r_j^s} + \frac{1}{2}}{n_{s,j-1}^0 + n_{s,j-1}^1 + 1}, \quad (\text{A.9})$$

415 where $n_{s,j-1}^a$ counts how many times symbol $a \in \{0, 1\}$ has appeared in context s before the j -th
 416 symbol of r^s .

417 The CTW mixture recursively combines the local KT model with the two child contexts:

$$P_w^s(r^s) = \begin{cases} P_e^s(r^s), & |s| = D, \\ \frac{1}{2}P_e^s(r^s) + \frac{1}{2}P_w^{0s}(r^{0s})P_w^{1s}(r^{1s}), & |s| < D. \end{cases} \quad (\text{A.10})$$

418 The one-step predictive probability at the root is then

$$\hat{p}_{\text{ctw},t} = \Pr_{\text{CTW}}(R_{t+1} = 1 \mid r_{1:t}) = \frac{P_w^\lambda(r_{1:t}1)}{P_w^\lambda(r_{1:t}0) + P_w^\lambda(r_{1:t}1)}. \quad (\text{A.11})$$

419 **Cold-start behavior.** At the beginning of a student sequence, most contexts are unseen. In that
 420 regime, the KT prior in (A.9) keeps $\hat{p}_{\text{ctw},t}$ close to 0.5, so $\hat{u}_{\text{ctw},t}$ in (A.8) stays close to one bit rather
 421 than becoming spuriously overconfident. As more evidence accumulates, longer contexts contribute
 422 through (A.10). For stationary finite-depth tree sources, the classical CTW per-symbol redundancy
 423 decays as $O(\log t/t)$ for a fixed source class [Willems et al., 1995], so the cold-start penalty vanishes
 424 with sequence length. In this paper we use that result only as a property of the reference source
 425 model: student learning itself is not assumed stationary, but the reference guarantee clarifies why
 426 CTW is a stable causal anchor.

Table 3: Dataset coverage used by the appendix. Support-interaction counts are reported for NIPS Task 3/4 and Algebra 2005.

Dataset	Protocol	Support interactions	Aligned evaluation points	Highest-band mass
NIPS Task 3/4	question-level	898,674	1,354,735	74.8%
Algebra 2005	question-level	598,954	464,725	55.6%
ASSIST2015	<i>concept = question</i>	–	672,960 for CTW+NN/KT; 673,200 for CTW before KT-side intersection	–

A.3 Mutual-information decomposition into RU and IG

For an item q , define the item-marginal entropy and the oracle conditional entropy by

$$H_q^{\text{marg}} := H(R_{t+1} \mid q_{t+1} = q), \quad H_q^{\text{cond}} := H(R_{t+1} \mid X_t, q_{t+1} = q). \quad (\text{A.12})$$

The theoretical remaining uncertainty and information gain are

$$\text{RU}^*(q) := H_q^{\text{cond}}, \quad \text{IG}^*(q) := H_q^{\text{marg}} - H_q^{\text{cond}}. \quad (\text{A.13})$$

Fact A.3 (RU/IG identity). *For every item q ,*

$$H_q^{\text{marg}} = \text{RU}^*(q) + \text{IG}^*(q), \quad \text{IG}^*(q) = I(R_{t+1}; X_t \mid q_{t+1} = q) \geq 0. \quad (\text{A.14})$$

Proof. By the definition of conditional mutual information,

$$I(R_{t+1}; X_t \mid q_{t+1} = q) = H(R_{t+1} \mid q_{t+1} = q) - H(R_{t+1} \mid X_t, q_{t+1} = q). \quad (\text{A.15})$$

Substituting (A.12) into (A.15) gives the second identity in (A.14); the first follows by rearrangement, and non-negativity is a standard property of mutual information. $\square$

The paper uses CTW to obtain operational plug-in diagnostics:

$$\text{RU}(q) := H_q^{\text{ctw}}, \quad \text{IG}(q) := H_q^{\text{marg}} - H_q^{\text{ctw}}. \quad (\text{A.16})$$

Because (A.16) replaces the oracle conditional entropy with an estimated one, small negative $\text{IG}(q)$ values may appear in finite samples.

Venn-style interpretation. It is useful to read (A.14) as a verbal Venn diagram. The full circle is the item-marginal entropy H_q^{marg} . The portion explained by student history is $\text{IG}^*(q)$, while the unexplained remainder is $\text{RU}^*(q)$. High IG therefore means that historical context removes a large fraction of the original uncertainty; high RU means that substantial uncertainty remains even after conditioning on the observable history.

Interpretation guide. If RU is high and IG is low, the item is a warning case: outcomes remain close to random under the observable protocol, which is consistent with guessing, ambiguous item wording, noisy labels, or missing explanatory variables. If IG is high, the benchmark still contains exploitable contextual structure, so stronger temporal, prerequisite, or item-context modeling is warranted.

B Appendix B: Data and Preprocessing Details

B.1 Dataset coverage

Table 3 summarizes the dataset coverage used in the appendix. The highest-band mass refers to the CTW-entropy band $[0.8, 1.0]$ used in the main benchmark-diagnosis figures.

Table 4: Implementation settings used by the appendix.

Component	Setting
CTW anchor	Binary response stream, maximum context depth $D = 6$, KT priors, causal one-step-ahead prediction.
Uncertainty bands	Five fixed CTW-entropy bands: $[0.0, 0.2)$, $[0.2, 0.4)$, $[0.4, 0.6)$, $[0.6, 0.8)$, $[0.8, 1.0]$.
CTW+NN probe	Fused logit correction $\sigma(\text{logit}(\hat{p}_{\text{ctw},t}) + f_\phi(X_{\leq t}, Q_{t+1}))$ used only as an anchor-side neural probe.
Synthetic sanity check	64 trials, sequence length 400, one-step warmup, CTW depth 6.
Item diagnostics	Minimum support 50 per item; diagnostic guide lines at $RU = 0.8$ and $IG = 0.1$.

B.2 Preprocessing protocol and alignment rules

The main benchmark-diagnosis figures use question-level preprocessing for NIPS Task 3/4 and Algebra 2005. ASSIST2015 is reported as an auxiliary consistency check under the convention *concept = question*, which makes the prediction unit compatible with the question-level protocol used by the two primary benchmarks. All reported comparisons use five-fold aligned held-out points: CTW-side anchor probabilities and task-facing KT predictions are joined only when they share the same fold, student/history position, item identifier, and held-out response.

The diagnostic stage operates on binary correctness labels and does not introduce a separate appendix-specific prediction task beyond the benchmark sequence protocol used by the main paper.

B.3 Knowledge-component mapping and Q-matrix status

The CTW anchor itself does not require a Q-matrix: it uses only the causal response history and the next queried item. The appendix does not introduce an additional inferred KC mapping layer.

C Appendix C: Implementation and Reproducibility

C.1 Implementation settings

Table 4 collects the implementation settings used by the appendix.

C.2 Training and evaluation details

All global metrics and local diagnostic summaries are reported as mean $\pm$ standard deviation over five folds. The CTW anchor and the derived bucketization are always computed out-of-fold. The shared aligned subset contains 1,354,735 points on NIPS Task 3/4, 464,725 points on Algebra 2005, and 672,960 points on ASSIST2015 under the convention *concept = question*.

The archived run exports used for the retained KT comparison table preserve fold-specific hyperparameter fields for each task-facing baseline. Table 5 summarizes representative recovered settings for the main baselines discussed in the paper. Each entry reports the set of values that appears across the five retained folds for that dataset, so multi-valued cells indicate genuine fold-level variation rather than post hoc averaging.

The same archive also preserves explicit training-length metadata for DIMKT on the two main benchmark datasets: batch size 64, num_steps 199, learning rate 10^{-3} , and weight decay 10^{-4} on both NIPS Task 3/4 and Algebra 2005, with dropout varying over $\{0.05, 0.1, 0.3\}$. We did not find optimizer, early-stopping, or wall-clock fields in the retained run exports, so those quantities are intentionally omitted from the submission version.

C.3 Compute-cost discussion

The CTW anchor is intentionally lightweight. For a fixed depth D , each prediction/update step touches only the suffix path of length at most D , so the per-step symbolic overhead is linear in D and small in practice. In our experiments we use $D = 6$, which is sufficient to separate the i.i.d. and

Table 5: Recovered fold-specific hyperparameter summaries for representative KT baselines in the archived run exports. Each cell reports the values appearing across the five retained folds for that dataset. Abbreviations: d for hidden width, d_{ff} for feed-forward width, heads for attention heads, B for transformer blocks, and mem for memory size.

Model	NIPS Task 3/4	Algebra 2005	ASSIST2015
DKT	emb 256, drop 0.3/0.5, lr 10^{-3}	emb 256, drop 0.1/0.3/0.5, lr 10^{-3}	emb 256, drop 0.05/0.3/0.5, lr 10^{-3}
AKT	$d = 64$, $d_{ff} = 64/256$, heads 4/8, $B = 2/4$, drop 0.1/0.3, lr 10^{-3}	$d = 64/256$, $d_{ff} = 256$, heads 4, $B = 4$, drop 0.05/0.1, lr 10^{-4}	$d = 64/256$, $d_{ff} = 64/256$, heads 4/8, $B = 4$, drop 0.05/0.1, lr $10^{-4}/10^{-5}$
simpleKT	$d = 64$, $d_{ff} = 256$, layers 2, $B = 4$, drop 0.3, lr 10^{-3}	$d = 256$, $d_{ff} = 64/256$, layers 2, $B = 4$, drop 0.05/0.1, lr $10^{-4}/10^{-5}$	$d = 256$, $d_{ff} = 256$, layers 2, $B = 4$, drop 0.05/0.1, lr $10^{-4}/10^{-5}$
stableKT	$d = 64$, $d_{ff} = 64/256$, layers 2, heads 4, drop 0.3, lr 10^{-3} , $r = 0.1/0.7/0.9/1.0$, $\gamma = 0.7/0.9/1.0/5.0$	$d = 256$, $d_{ff} = 64/256$, layers 2, heads 4, drop 0.05/0.1, lr $10^{-4}/10^{-5}$, $r = 0.5/0.9/5.0$, $\gamma = 0.1/0.3/0.5/5.0$	$d = 256$, $d_{ff} = 64/256$, layers 2, heads 4, drop 0.05/0.1, lr 10^{-5} , $r = 0.5/1.0/2.0$, $\gamma = 0.1/0.5/0.7$
Deep-IRT	$\dim_s = 256$, mem 32/64, drop 0.05/0.3, lr 10^{-3}	$\dim_s = 256$, mem 32, drop 0.05/0.1, lr 10^{-3}	$\dim_s = 64$, mem 32, drop 0.05/0.1, lr 10^{-3}
ATKT	skill 64, ans 64/256, hidden 256, attn 64/256, drop 0.05/0.1, lr $10^{-4}/10^{-3}$	skill 64, ans 64, hidden 256, attn 256, drop 0.05/0.1/0.5, lr 10^{-3}	skill 64, ans 64/256, hidden 256, attn 64/256, drop 0.05, lr $10^{-4}/10^{-3}$

Table 6: Cross-benchmark diagnostic contrasts. Positive ΔACC means CTW+NN improves over CTW; negative ΔBrier means the proper score improves.

Dataset	Highest CTW band mass	ΔACC in highest CTW band	ΔBrier in highest CTW band	Refined-coordinate note
NIPS Task 3/4	74.8%	+0.0886	-0.0378	Under refined repartitioning, the peak CTW+NN gain shifts to the upper-middle bucket at +0.0868.
Algebra 2005	55.6%	+0.0283	-0.0681	Under refined repartitioning, the largest CTW+NN gain appears in the fourth bucket at +0.0335, while gains remain positive in the highest bucket.

485 persistent synthetic regimes while keeping the causal anchor inexpensive. This is one of the main
486 practical advantages of the framework: the uncertainty analysis adds a modest symbolic pass on top
487 of predictions that are already being generated for KT evaluation, whereas the expensive part of the
488 pipeline remains the training of the task-facing neural KT models.

489 D Appendix D: Additional Experimental Results

490 D.1 Cross-benchmark diagnostic differences

491 Table 6 summarizes the strongest cross-benchmark contrasts reported in the paper. The main point is
492 not only that the high-entropy region exists, but that it carries different mass and different residual
493 headroom across datasets.

494 The table reinforces the same diagnosis from two angles. On NIPS Task 3/4, the highest-entropy
495 region dominates the benchmark mass and still admits the largest accuracy gain over CTW, so the
496 benchmark is clearly not saturated in its difficult regime. On Algebra 2005, the hard region is smaller
497 but still substantial, and the largest proper-score improvement also appears there. Thus RU/IG and
498 band-level gains should be read together: high entropy can still hide exploitable structure, but the
499 gain pattern differs by dataset.

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